Lab 5, Act 1: Finding the Peak

What detection probability do the survey data support best?

NRES 746

Answer key. Answers are in magenta.

The scenario

A rare frog in a Sierra Nevada spring complex calls only on some nights, so a survey can miss it even when it's present at a site. To estimate how often a single visit detects it, you visit three distinct headwaters where the species is *known to be present*, ten times each. On each visit you listen for 5 minutes and record whether you heard at least one call. Here are your observed measurements:

Wetland	Visits	Visits with a detection
A	10	3
B	10	2
C	10	7

Assume every wetland has the same single-visit detection probability p, and that visits are independent.

What value of p do these data support best, and can we find it without running a brute-force grid search?

Please keep your laptop closed for this section (calculator/phone is okay!).

Part 1: The likelihood (individual, ~15 min)

1a. Write the likelihood of all three wetlands' data, $\mathcal{L}_3(p)$, using product notation.

$$\mathcal{L}_3(p) = \prod_{i=1}^{3} \binom{10}{k_i} p^{k_i}(1-p)^{10-k_i} = \binom{10}{3}\binom{10}{2}\binom{10}{7} p^{12}(1-p)^{18}$$

1b. Take the log and simplify $\ell_3(p) = \log \mathcal{L}_3(p)$ into a sum, as far as you can.

$$\ell_3(p) = \sum_{i=1}^{3} \left[\log \binom{10}{k_i} + k_i \log p + (10 - k_i)\log(1-p) \right] = C + 12\log p + 18\log(1-p)$$

with $C = \log(120 \cdot 45 \cdot 120) \approx 13.38$.

1c. Your answer to 1b includes binomial coefficients such as $\binom{10}{3}$. If you left them out, would that change which value of p maximizes $\ell_3(p)$? Why or why not?

No. They add a constant that doesn't depend on p, which shifts the whole curve up or down without moving its peak.

Part 2: Find the peak (individual, ~20 min)

2a. Find the derivative with respect to p of either $\mathcal{L}_3(p)$ or $\ell_3(p)$, your choice (both peak at the same p). In general, the log-likelihood is easier to differentiate.

$$\frac{d\ell_3}{dp} = \frac{12}{p} - \frac{18}{1-p}$$

2b. Set your derivative equal to zero and solve for $\hat{p}$, the maximum likelihood estimate. Try to do this on your own first, but you can get help by looking at the Bolker book (beginning of Ch. 6).

$\frac{12}{p} = \frac{18}{1-p} \Rightarrow 12(1-p) = 18p \Rightarrow 12 = 30p$

$\hat{p} = 12/30 = 0.4$

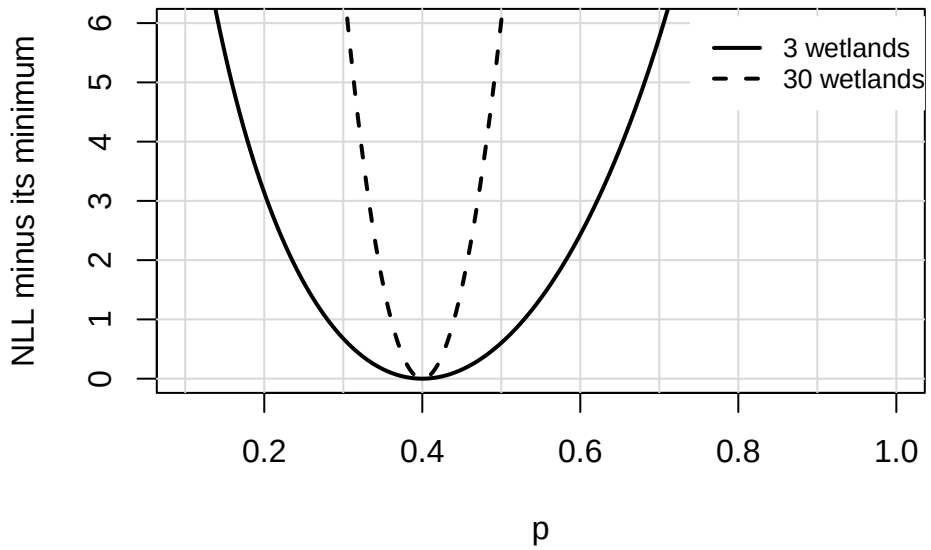
2c. Now suppose you surveyed n wetlands, each visited N times, with k_i detections at wetland i. Write a general formula for $\hat{p}$. Does it make sense?

$\hat{p} = \frac{\sum_{i=1}^{n} k_i}{nN}$

Yes: total detections divided by total visits, the observed detection rate.

Part 3: How sure are we? (individual, ~10 min)

A colleague ran the same survey at 30 spring complexes where the species is known to be present (10 visits each) and heard the frog on 120 of the 300 visits. The figure shows the negative log-likelihood (NLL) curve for each survey, each shifted down so its minimum sits at zero.



3a. Do the two surveys give the same $\hat{p}$? Which survey pins p down more tightly, and what feature of the curve shows it?

Yes, both give $\hat{p} = 0.4$ ($120/300 = 12/30$). The 30-wetland survey: its curve is much narrower and steeper around the minimum.

Group discussion and revision (~10 min)

Compare your answers with your group, especially your derivative in 2a and your answer to 3a, and revise as needed. Record any changes you made, new insights, and any "muddiest points" or topics you want to review.

Open-ended.